Network Modelling Assignment 1

Group 19

2026-03-20

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1 Task 1

1.1 Task 1.1

There are 22 pupils, so the total number of potential ties amongst them is
$n(n-1) = 22 * 21 = 462$.
We are given that there are $m = 134$ ties in the network, therefore the probability p of observing a tie is
$p = \frac{m}{n(n-1)} = \frac{134}{462} = 0.290$

1.2 Task 1.2

There is a total of $\binom{n}{2} = \binom{22}{2} = 231$ possible dyads. The probability of observing a mutual dyad is
$p_M = \frac{M}{231} = 0.182$
The probability of observing an asymmetric dyad is
$p_A = \frac{A}{231} = 0.217$
The probability of observing a null dyad is
$p_N = \frac{N}{231} = 0.602$

1.3 Task 1.3

Assuming independence:
The probability of a mutual dyad is:
$p'_M = P(X_{ij} = 1)P(X_{ji} = 1) = p^2 = 0.084$

The probability of a null dyad is:
$p'_N = P(X_{ij} = 0)P(X_{ji} = 0) = (1-p)^2 = 0.412$

The probability of an asymmetric dyad is:
$p'A = P(X_{ij} = 1)P(X_{ji} = 0) + P(X_{ij} = 0)P(X_{ji} = 1) = 2p(1-p) = 0.504$

1.4 Task 1.4

The probability of mutual dyads observed is much higher than when assuming independence. For asymmetric dyads, the probability observed is almost half compared to when assuming independence. For null dyads, the observed probability is again higher than when assuming independence. This suggests dependence between dyads, and therefore it would not be reasonable to assume independence.

1.5 Task 1.5

If we have a model that assumes that all nodes have an outdegree of 9 (d=9), then
$P(X_{ij} = 1) = \frac{9}{21} = 0.429$
Then, assuming independence the probability of a mutual tie is:
$p''_M = P(X_{ij} = 1)^2 = \frac{9}{49} = 0.184$
This mutual tie probability of 0.184 is very close to the observed probability that we calculated in part (2) which was 0.182. While this could suggest that this could be a good approximation, the model is quite unreasonable and independence still might not be a good assumption, since we fix $d = 9$. This model has a density much higher than the observed network, and it only reaches the correct probability of mutual ties, since it has way more ties overall (22*9=198 ties) than the observed network (134 observed ties). This model basically increases both the tie probability and the total number of ties, and these two errors happen to cancel out for the mutual dyad probability. Fixing d=9 for all nodes is an unrealistic constraint and the model fails to capture the actual degree heterogeneity in the network. The coincidential match with p_M does not imply the model is a good fit for the network as a whole.

1.6 Task 1.6

```
set.seed(161)
library(sna)
library(network)

obsMat <- as.matrix(read.csv2("classrooms/10_W1.csv", header = TRUE, row.names = 1, check.names = FALSE)


#data cleaning
na_nodes <- which(rowSums(is.na(obsMat)) == ncol(obsMat))

obsMat_clean <- obsMat[-na_nodes, -na_nodes, drop = FALSE]



cguRec <- cug.test(obsMat_clean, grecip, cmode = "edges", reps=5000)
cugInd <- cug.test(obsMat_clean, centralization, cmode="edges", FUN.arg=list(FUN=degree, cmode="indegree
cugTrans <- cug.test(obsMat_clean, gtrans, cmode = "dyad.census", reps=5000)
```

1.6.1 Task 1.6.1

cguRec:
The null hypothesis is H_0 : The observed reciprocity is what we would expect in a random network with the same number of edges.
The alternative hypothesis is H_1 : The observed reciprocity is different than what we would expect in a random network with the same number of edges.

cugInd:
The null hypothesis is H_0 : The observed indegree centralization is what we would expect in a random network with the same number of edges.
The alternative hypothesis is H_1 : The observed indegree centalization is different than what we would expect in a random network with the same number of edges.

cugTrans:
The null hypothesis is H_0 : The observed transitivity is what we would expect in a random network with

the same number of edges.
The alternative hypothesis is H_1 : The observed transitivity is different than what we would expect in a random network with the same number of edges.

1.6.2 Task 1.6.2

For the **cguRec** test we get the following results:

```
##
## Univariate Conditional Uniform Graph Test
##
## Conditioning Method: edges
## Graph Type: digraph
## Diagonal Used: FALSE
## Replications: 5000
##
## Observed Value: 0.7736842
## Pr(X>=Obs): 0
## Pr(X<=Obs): 1
```

The observed value is 0.774 and the p-values are $Pr(X >= Obs) = 0$, $Pr(X <= Obs) = 1$. This means that the observed reciprocity is significantly larger than expected. In fact, none of the simulated models produced reciprocity as high as the observed one. Therefore, there is a strong tendency for reciprocated ties in the network.

For the **cugInd** test we get the following results:

```
##
## Univariate Conditional Uniform Graph Test
##
## Conditioning Method: edges
## Graph Type: digraph
## Diagonal Used: FALSE
## Replications: 5000
##
## Observed Value: 0.2022161
## Pr(X>=Obs): 0.7712
## Pr(X<=Obs): 0.6252
```

The observed value is 0.202 and the p-values are $Pr(X >= Obs) = 0.7712$, $Pr(X <= Obs) = 0.6252$. This means that the observed indegree centralization in not significantly different from what we would expect in a random network.

For the **cugTrans** test we get the following results:

```
##
## Univariate Conditional Uniform Graph Test
##
## Conditioning Method: dyad.census
## Graph Type: digraph
## Diagonal Used: FALSE
## Replications: 5000
##
```

```
## Observed Value: 0.5360195
## Pr(X>=Obs): 0
## Pr(X<=Obs): 1
```

The observed value is 0.536 and the p-values are $Pr(X >= Obs) = 0$, $Pr(X <= Obs) = 1$. This means that the observed transitivity is significantly larger than expected. In fact, none of the simulated models produced transitivity as high as the observed one. Therefore, there is a strong tendency for transitive ties in the network.

2 Data Cleaning

Data Cleaning Note: Our data processing followed a standardized procedure to ensure consistency between relational and attribute datasets. First, the network matrices were cleaned by identifying and removing nodes that contained only missing values, ensuring the integrity of the weight matrices. Second, we applied listwise deletion to the attribute data to ensure that only students with complete records across all relevant variables were included in the estimation. Finally, we synchronized the two datasets by matching student identifiers, retaining only the intersection of students present in both the cleaned attributes and the networks to maintain a valid square weight matrix for the lnam estimation.

```r
library(sna)
library(network)

input_folder = "/Users/duartealbuquerque/Desktop/ETH-Switzerland/ETH_Classes/S26/Network Modeling/Proje
setwd(folder_path)
output_folder <- "/Users/duartealbuquerque/Desktop/ETH-Switzerland/ETH_Classes/S26/Network Modeling/Pro

file_list = list.files(path = folder_path, pattern = "*.csv")

for (file_name in file_list) {

  file_path <- file.path(input_folder, file_name)
  obsMat <- as.matrix(read.csv2(file_path, header = TRUE, row.names = 1, check.names = FALSE))
  na_indices <- which(rowSums(is.na(obsMat)) == ncol(obsMat))

  if (length(na_indices) > 0) {
    obsMat_clean <- obsMat[-na_indices, -na_indices, drop = FALSE]
  } else {
    obsMat_clean <- obsMat
  }

  output_path <- file.path(output_folder, paste0("cleaned_", file_name))
  write.csv2(obsMat_clean, file = output_path, row.names = TRUE)

}

input_file_attr = "/Users/duartealbuquerque/Desktop/ETH-Switzerland/ETH_Classes/S26/Network Modeling/Pr
attr_data <- read.csv2(input_file_attr,
                       header = TRUE,
                       row.names = 1,
                       check.names = FALSE,
                       na.strings = c("", " ", "NA"))
```

```r
attr_clean <- na.omit(attr_data)


output_path_attr <- file.path(output_folder, "cleaned_attr.csv")
write.csv2(attr_clean, file = output_path_attr, row.names = TRUE)
```

3 Task 2

3.1 Setup

```r
library(sna)
library(network)

set.seed(161)

# Path to the cleaned adjacency matrix CSVs (relative to this file)
data_dir  <- "cleaned_csv"
# Attribute file (comma-separated, produced by cleaning.R via write.csv)
attr_file <- file.path(data_dir, "cleaned_attr.csv")

classrooms <- c("01", "02", "03", "04", "05", "09", "10", "12")
```

3.2 Helper functions

We begin by defining some helper functions. `build_data` is our cleaning pipeline, `run_qap` is a cleaner expression of the `netlm` function, and `tidy_netlm` gets a tidy dataframe from the `netlm` fit.

```r
# Load an adjacency matrix from a cleaned CSV.
# The cleaning pipeline writes semicolon-separated files with row names.
load_adj <- function(file) {
  mat_raw <- read.csv(file, sep = ";", row.names = 1,
                      header = TRUE, check.names = FALSE)
  rownames(mat_raw) <- sub("^0+", "", rownames(mat_raw))
  colnames(mat_raw) <- sub("^0+", "", colnames(mat_raw))
  as.matrix(mat_raw)
}

# Build cleaned predictor matrices for a given classroom ID string
# (e.g. "10" -> cleaned_10_W1.csv / cleaned_10_W2.csv).
build_data <- function(classroom) {
  f1 <- file.path(data_dir, paste0("cleaned_", classroom, "_W1.csv"))
  f2 <- file.path(data_dir, paste0("cleaned_", classroom, "_W2.csv"))

  W1_raw <- load_adj(f1)
  W2_raw <- load_adj(f2)

  # Restrict to students present in both waves
  ids_common <- intersect(rownames(W1_raw), rownames(W2_raw))
  W1 <- W1_raw[ids_common, ids_common]
```

```r
  W2 <- W2_raw[ids_common, ids_common]
  storage.mode(W1) <- "numeric"
  storage.mode(W2) <- "numeric"

  # Flag nodes with any NA in their nomination row in either wave
  # (partial non-response; all-NA rows were already removed by cleaning.R)
  missing_net <- rowSums(is.na(W1)) > 0 | rowSums(is.na(W2)) > 0

  # Match attributes; students absent from attr_raw get an all-NA row
  attr_sub <- attr_raw[match(ids_common, as.character(attr_raw$studentID)), ]

  # Flag nodes with any missing required covariate
  missing_attr <- is.na(attr_sub$literacy_end) |
                  is.na(attr_sub$HISEI) |
                  is.na(attr_sub$gender)

  keep       <- !(missing_net | missing_attr)
  ids_clean  <- ids_common[keep]
  W1_c       <- W1[keep, keep]
  W2_c       <- W2[keep, keep]
  attr_c     <- attr_sub[keep, ]

  n   <- nrow(W1_c)
  lit <- as.numeric(attr_c$literacy_end)
  gen <- attr_c$gender
  his <- as.numeric(attr_c$HISEI)

  # Predictor matrix: receiver literacy  X[i,j] = literacy_end of j
  X_reclit <- matrix(rep(lit, each = n), nrow = n, ncol = n)

  # Predictor matrix: same-gender dyad  X[i,j] = 1 if gender_i == gender_j
  X_gender <- outer(gen, gen, FUN = function(a, b) as.integer(a == b))

  # Predictor matrix: sender HISEI  X[i,j] = HISEI of i
  X_hisei <- matrix(rep(his, times = n), nrow = n, ncol = n)

  # Zero diagonals (no self-nominations)
  diag(W1_c) <- diag(W2_c) <- 0
  diag(X_reclit) <- diag(X_gender) <- diag(X_hisei) <- 0

  list(Y = W2_c, X_w1 = W1_c, X_reclit = X_reclit,
       X_gender = X_gender, X_hisei = X_hisei,
       ids = ids_clean, n = n)
}

# Run MR-QAP regression (netlm with QAP permutation)
run_qap <- function(dat, reps = 5000) {
  netlm(dat$Y,
        list(dat$X_w1, dat$X_reclit, dat$X_gender, dat$X_hisei),
        reps    = reps,
        mode    = "digraph",
        nullhyp = "qap")
}
```

```r
# Extract a tidy result data-frame from a netlm fit
tidy_netlm <- function(fit, classroom) {
  data.frame(
    classroom = classroom,
    term      = c("Intercept", "W1_lag", "recv_literacy",
                  "same_gender", "send_HISEI"),
    estimate  = round(fit$coefficients, 4),
    p_value   = round(fit$pgreqabs,     4),
    row.names = NULL
  )
}
```

3.3 Load attribute data

```r
attr_raw <- read.csv(attr_file, sep = ";", stringsAsFactors = FALSE)
attr_raw$studentID <- as.character(attr_raw$X)
```

3.4 Task 2.1 - Simple QAP for Classroom 10

We first assess whether friendship nominations in Wave 2 are predicted by the same nominations in Wave 1, using a simple QAP regression with the lagged network as the only predictor:

$$Y_{ij}^{W2} = \beta_0 + \beta_1 \cdot Y_{ij}^{W1} + \varepsilon_{ij}$$

The **null hypothesis** is that the observed association between $W1$ and $W2$ is weak enough such that it could reasonably arise by chance under random permutation of the network.

```r
dat10 <- build_data("10")
cat("Classroom 10 - nodes retained after cleaning:", dat10$n, "\n")
```

```
## Classroom 10 - nodes retained after cleaning: 16
```

```r
fit10_simple <- netlm(dat10$Y, list(dat10$X_w1),
                      reps = 5000, mode = "digraph", nullhyp = "qap")
summary(fit10_simple)
```

```
##
## OLS Network Model
##
## Residuals:
##          0%         25%         50%         75%        100%
## -0.92857143 -0.41025641  0.07142857  0.58974359  0.58974359
##
## Coefficients:
##             Estimate  Pr(<=b) Pr(>=b) Pr(>=|b|)
## (intercept) 0.4102564 1       0       0
```

```
## x1          0.5183150 1       0       0
##
## Residual standard error: 0.4266 on 238 degrees of freedom
## Multiple R-squared: 0.253    Adjusted R-squared: 0.2498
## F-statistic:  80.6 on 1 and 238 degrees of freedom, p-value: 1.11e-16
##
##
## Test Diagnostics:
##
##  Null Hypothesis: qap
##  Replications: 5000
##  Coefficient Distribution Summary:
##
##        (intercept)       x1
## Min        4.78552 -4.48272
## 1stQ       7.23656 -1.01668
## Median     7.87402  0.08226
## Mean       7.93081  0.01243
## 3rdQ       8.58530  0.90636
## Max       11.73186  5.63351
```

Interpretation: The coefficient on the lagged Wave-1 network is positive ($\beta_1 \approx 0.52$) and significant at the 5% level, which means we can reject the null. Friendships present at Wave 1 are strongly predictive of ties at Wave 2. This is what we'd expect to see intuitively: friendships are stable across time (relatively, at least–this is high school, after all). In network science terminology, we could say that our network has some intertia or tie-persistence.

3.5 Task 2.2 - Operationalisation of Hypotheses

The full MR-QAP model is:

$$Y_{ij}^{W2} = \beta_0 + \beta_1 Y_{ij}^{W1} + \beta_2 X_{ij}^{\text{reclit}} + \beta_3 X_{ij}^{\text{gender}} + \beta_4 X_{ij}^{\text{HISEI}} + \varepsilon_{ij}$$

We add three dyadic predictor matrices to test the following hypotheses:

Hypothesis	Operationalisation	Entry X_{ij}
i. High-literacy receivers get *fewer* nominations	**Receiver literacy**	literacy_end_j
ii. Same-gender pairs more likely to nominate	**Same gender**	$\mathbf{1}[\text{gender}_i = \text{gender}_j]$
iii. High-HISEI senders nominate *more*	**Sender HISEI**	HISEI_i

We tried to choose the simplest possible representation of each effect.

- *Hypothesis i* is an **alter (receiver) effect**: variation in how often a node is *chosen* is captured by a column-constant matrix whose entry in column j equals j's literacy score.
- *Hypothesis ii* is a **homophily / dyadic covariate**: a binary indicator equals 1 when sender and receiver share the same gender. No other encoding would make sense here.

- *Hypothesis iii* is an **ego (sender) effect**: variation in how many nominations a node *sends* is captured by a row-constant matrix whose entry in row i equals i's HISEI score.

3.6 Task 2.3 - MR-QAP Estimation for Classroom 10

We re-run the model…

```
fit10 <- run_qap(dat10, reps = 5000)
summary(fit10)
```

```
##
## OLS Network Model
##
## Residuals:
##          0%         25%         50%         75%        100%
## -1.00320768 -0.28045459 -0.04716163  0.30131079  0.82294805
##
## Coefficients:
##             Estimate     Pr(<=b) Pr(>=b) Pr(>=|b|)
## (intercept) -0.064927519 0.3890  0.6110  0.7822
## x1           0.343241791 1.0000  0.0000  0.0000
## x2           0.002573003 0.9374  0.0626  0.1198
## x3           0.342849435 1.0000  0.0000  0.0000
## x4           0.004437190 0.9042  0.0958  0.1800
##
## Residual standard error: 0.3943 on 235 degrees of freedom
## Multiple R-squared: 0.3699   Adjusted R-squared: 0.3592
## F-statistic: 34.49 on 4 and 235 degrees of freedom, p-value:     0
##
##
## Test Diagnostics:
##
##  Null Hypothesis: qap
##  Replications: 5000
##  Coefficient Distribution Summary:
##
##        (intercept)        x1        x2        x3        x4
## Min      -6.891082 -4.851617 -4.396652 -3.937945 -5.628444
## 1stQ     -1.327244 -0.900495 -1.052702 -0.731555 -1.328309
## Median   -0.002148  0.006511 -0.026316 -0.016150 -0.091162
## Mean     -0.003517  0.008707 -0.001461  0.019289 -0.010415
## 3rdQ      1.376816  0.914620  1.013780  0.728328  1.273687
## Max       6.784571  4.588835  4.741268  5.656062  6.264470
```

…and obtain:

```
tidy10 <- tidy_netlm(fit10, "10")
knitr::kable(tidy10, digits = 4,
             caption = "MR-QAP coefficients - Classroom 10")
```

Table 2: MR-QAP coefficients - Classroom 10

classroom	term	estimate	p_value
10	Intercept	-0.0649	0.7822
10	W1_lag	0.3432	0.0000
10	recv_literacy	0.0026	0.1198
10	same_gender	0.3428	0.0000
10	send_HISEI	0.0044	0.1800

3.7 Interpretation

The QAP permutation test randomly shuffles row and column labels of the response matrix to obtain the null distribution of each coefficient. The reported p-value (`pgreqabs`) is the proportion of permuted coefficients at least as extreme (in absolute value) as the observed one.

Term	Expected direction	Observed	Supported?
`W1_lag`	positive (stability)	0.343 ($p = 0$)	Yes
`recv_literacy`	negative	0.003 ($p = 0.12$)	No
`same_gender`	positive	0.343 ($p = 0$)	Yes
`send_HISEI`	positive	0.004 ($p = 0.18$)	No

We now comment on each hypothesis:

- *Lagged network (W1):* Even when controlling for three further variables, we find that the tie persistence property is well-supported at the 5%-level. In fact, of the correlates investigated, being friends in a prior period is the strongest determinant of current friendship together with gender (the effect strength is equal).
- *Receiver literacy.* We were asked to test the one-sided hypothesis that literacy scores negatively correlate with friendship nominations. That means we would have expected a negative coefficient. We found no evidence of this: the parameter estimate was close to zero and not supported at the 5% level. We fail to reject the null.
- *Same gender.* A positive, significant coefficient supports the gender-homophily hypothesis: students prefer being friends with same-gender peers. Interestingly, the effect strength (and significance) is very close to the tie persistence hypothesis.
- *Sender HISEI.* A positive, significant coefficient would indicate that students from higher socioeconomic backgrounds tend to send more friendship nominations. As with the literacy hypothesis, we find no conclusive information. The estimate is close to zero and the significance is far too low to reject at the 5% level. (In fact, a p-value of 0.18 indicates that we'd expect to see such an effect in every fifth simulated network.)

3.8 Task 2.4 - All Remaining Classrooms

We re-run our analysis and provide output tables. In broad strokes, our findings remain broadly the same. There are some aberrations–we comment on these in 6.4.

```r
all_results <- list()
all_tidy    <- list()

for (cl in classrooms) {
  dat <- tryCatch(build_data(cl), error = function(e) NULL)
  if (is.null(dat) || dat$n < 5) next

  fit              <- run_qap(dat, reps = 5000)
  all_results[[cl]] <- fit
  all_tidy[[cl]]    <- tidy_netlm(fit, cl)

  cat(sprintf("\n## Classroom %s (n = %d)\n\n", cl, dat$n))
  print(knitr::kable(all_tidy[[cl]], digits = 4,
                     caption = sprintf("Classroom %s", cl)))
  cat("\n")
}
```

3.9 Classroom 01 (n = 23)

Table 4: Classroom 01

classroom	term	estimate	p_value
01	Intercept	0.4302	0.0240
01	W1_lag	0.4214	0.0000
01	recv_literacy	-0.0006	0.6586
01	same_gender	0.2150	0.0002
01	send_HISEI	-0.0040	0.1134

3.10 Classroom 02 (n = 19)

Table 5: Classroom 02

classroom	term	estimate	p_value
02	Intercept	0.3837	0.0326
02	W1_lag	0.4149	0.0000
02	recv_literacy	-0.0027	0.0714
02	same_gender	0.1495	0.0068
02	send_HISEI	-0.0015	0.5198

3.11 Classroom 03 (n = 17)

Table 6: Classroom 03

classroom	term	estimate	p_value
03	Intercept	0.0505	0.5150
03	W1_lag	0.2843	0.0000
03	recv_literacy	0.0002	0.8952

classroom	term	estimate	p_value
03	same_gender	0.2902	0.0002
03	send_HISEI	0.0001	0.9186

3.12 Classroom 04 (n = 18)

Table 7: Classroom 04

classroom	term	estimate	p_value
04	Intercept	0.3688	0.1488
04	W1_lag	0.4367	0.0000
04	recv_literacy	0.0006	0.5984
04	same_gender	0.0706	0.2346
04	send_HISEI	-0.0026	0.5140

3.13 Classroom 05 (n = 24)

Table 8: Classroom 05

classroom	term	estimate	p_value
05	Intercept	0.3339	0.0214
05	W1_lag	0.5246	0.0000
05	recv_literacy	-0.0003	0.8326
05	same_gender	0.1506	0.0004
05	send_HISEI	-0.0034	0.0590

3.14 Classroom 09 (n = 15)

Table 9: Classroom 09

classroom	term	estimate	p_value
09	Intercept	-0.1049	0.2986
09	W1_lag	0.6188	0.0000
09	recv_literacy	0.0010	0.4726
09	same_gender	0.1460	0.0184
09	send_HISEI	0.0021	0.0498

3.15 Classroom 10 (n = 16)

Table 10: Classroom 10

classroom	term	estimate	p_value
10	Intercept	-0.0649	0.7914
10	W1_lag	0.3432	0.0000

classroom	term	estimate	p_value
10	recv_literacy	0.0026	0.1146
10	same_gender	0.3428	0.0000
10	send_HISEI	0.0044	0.1866

3.16 Classroom 12 (n = 17)

Table 11: Classroom 12

classroom	term	estimate	p_value
12	Intercept	0.4984	0.0134
12	W1_lag	0.5361	0.0000
12	recv_literacy	0.0007	0.7842
12	same_gender	0.0270	0.7066
12	send_HISEI	-0.0031	0.2916

```
all_tidy_df <- do.call(rbind, all_tidy)
```

3.17 Task 2.5 - Cross-Classroom Summary and Generalisation

Overall, we have found the following significant effects:

```
sig_counts <- tapply(all_tidy_df$p_value < 0.05,
                     all_tidy_df$term, sum)
print(sig_counts)
```

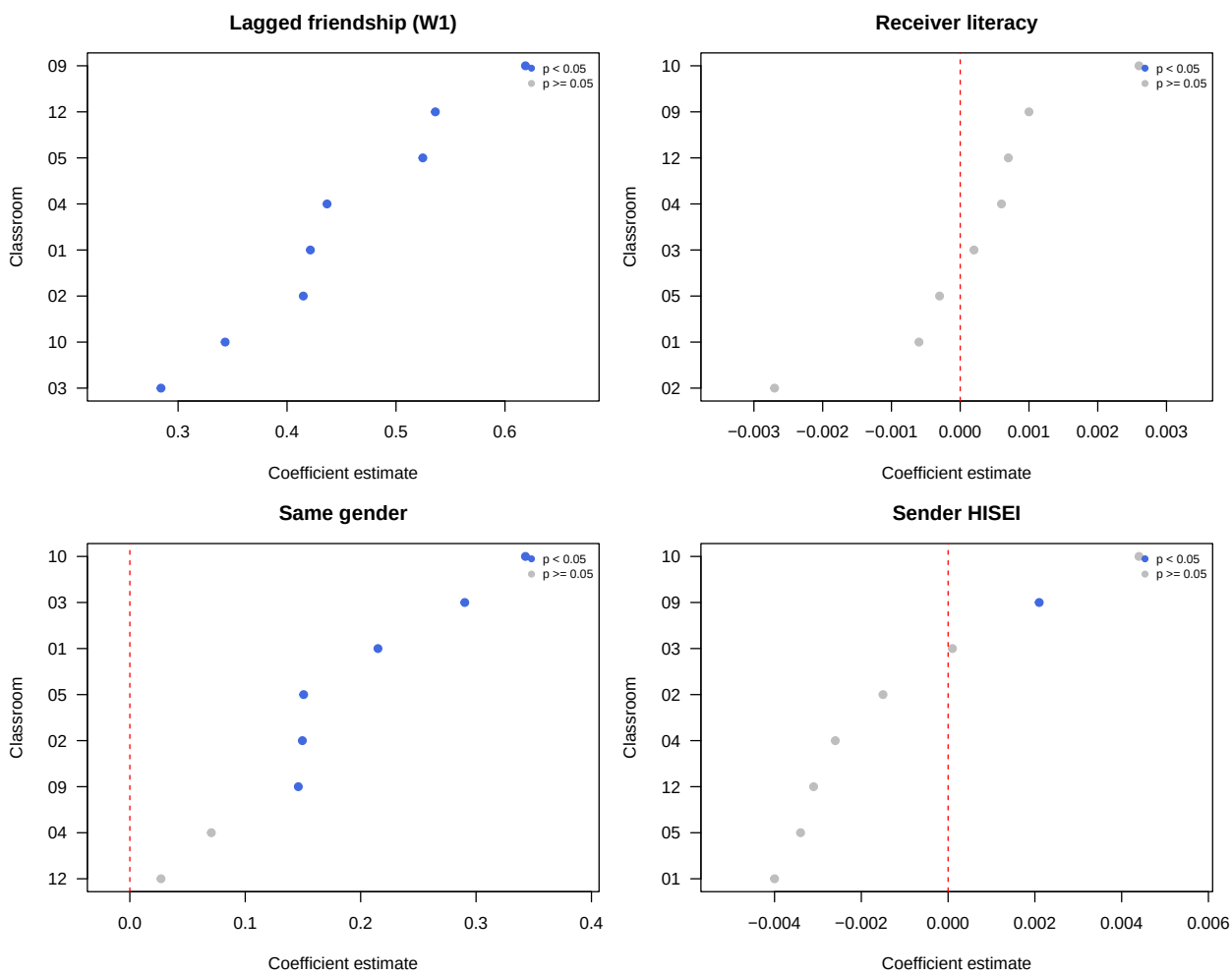
```
##     Intercept recv_literacy   same_gender    send_HISEI        W1_lag
##             4             0             6             1             8
```

3.18 Caterpillar plots

We now visualize our data via caterpillar plots. Each panel shows the estimated coefficient for one predictor across all classrooms, sorted by effect size. Blue points are significant at the $p < 0.05$ level.

```
terms_plot <- c("W1_lag", "recv_literacy", "same_gender", "send_HISEI")
term_labels <- c(
  W1_lag        = "Lagged friendship (W1)",
  recv_literacy = "Receiver literacy",
  same_gender   = "Same gender",
  send_HISEI    = "Sender HISEI"
)

par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
for (trm in terms_plot) {
  sub  <- all_tidy_df[all_tidy_df$term == trm, ]
```

```r
  sub  <- sub[order(sub$estimate), ]
  cols <- ifelse(sub$p_value < 0.05, "royalblue", "grey")

  rng <- range(sub$estimate)
  pad <- diff(rng) * 0.15

  plot(sub$estimate, seq_len(nrow(sub)),
       xlab  = "Coefficient estimate",
       ylab  = "Classroom",
       main  = term_labels[trm],
       pch   = 19,
       col   = cols,
       yaxt  = "n",
       xlim  = c(rng[1] - pad, rng[2] + pad))
  axis(2, at = seq_len(nrow(sub)), labels = sub$classroom, las = 1)
  abline(v = 0, lty = 2, col = "red")
  legend("topright",
         legend = c("p < 0.05", "p >= 0.05"),
         col    = c("royalblue", "grey"),
         pch    = 19, cex = 0.75, bty = "n")
}
```



3.19 Discussion

Friendship stability (`W1_lag`): The lagged wave-1 network has a positive and statistically significant coefficient in nearly every classroom. This is our most robust finding: friendships formed at the start of the school year largely persist until the end, consistent with what the literature describes as *tie persistence* or *network inertia*.

Receiver literacy (Hypothesis i): The direction and significance of this coefficient varies across classrooms, suggesting the relationship between academic achievement and popularity is not uniform. In some classrooms a negative coefficient (fewer nominations for high-literacy students) appears; in others the effect is negligible or reversed. Hypothesis i therefore receives only *weak and inconsistent* support.

Same gender (Hypothesis ii): Gender homophily—the tendency to nominate same-gender peers more often—is present in several classrooms with a positive coefficient. However, the effect is not universally significant (e.g. classrooms 2 and 14). The reasons for this would merit further research. Are there other variables that describe when (and, if we took a causal view, why) gender homophily breaks down?

Sender HISEI (Hypothesis iii): The association between socioeconomic status and outgoing friendship nominations is modest and inconsistent. High-HISEI students do not systematically nominate more friends in every setting, suggesting that this effect, if present, depends on classroom composition and context. We do find that classroom 9's socioeconomic hypothesis is (just barely) significant at the 5% level. We are cautious and chalk this up to multiple testing: a 5% significance means that we'd expect a significant effect for every 20 hypotheses by pure chance. We have tested far more than 20. Further, the coefficient value (≈ 0.0021) is hardly inspiring.

Generalisability: The findings rest on only eight classrooms from a single Czech school cohort, limiting external validity. The heterogeneity in effect sizes and significance levels across classrooms underscores that adolescent peer-network dynamics are context-dependent. Classroom composition (gender ratio, HISEI distribution, class size) likely moderates individual effects. A broader, multi-school sample would be required for generalisation. Differences across classrooms for the gender and literacy predictors could reflect real variation in how these characteristics shape friendship selection, or they could reflect sampling variability given the relatively small number of students per classroom.

4 Task 3

4.1 Task 3.1 - Basic NAM (Social influence only)

The Network Autocorrelation Model (NAM) investigates whether a student's literacy score is influenced by the scores of their friends (social influence). For Classroom 10, we utilize the friendship network from Wave 2 and the literacy scores at the end of the year (literacy_end).

```
attr_data <- read.csv('cleaned_csv/cleaned_attr_lit_end.csv', sep=";", header=TRUE)
network_w2 <- read.csv('cleaned_csv/cleaned_10_W2.csv', sep=";", header=TRUE, row.names=1)

classroom_10_attr <- attr_data[attr_data$classroomID == 10, ]
network_matrix <- as.matrix(network_w2)

student_ids_in_network <- rownames(network_matrix)
literacy_scores <- as.numeric(classroom_10_attr$literacy_end[match(student_ids_in_network, as.character

valid_indices <- !is.na(literacy_scores)
network_matrix_clean <- network_matrix[valid_indices, valid_indices]
literacy_scores_clean <- literacy_scores[valid_indices]
```

```r
nam_model <- lnam(y = literacy_scores_clean, W1 = network_matrix_clean)

summary(nam_model)
```

```
##
## Call:
## lnam(y = literacy_scores_clean, W1 = network_matrix_clean)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##   5.13  35.25  48.72  66.99  84.62
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## rho1.1  0.07330    0.01058   6.928 4.26e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    30.39      26.48
##
## Goodness-of-Fit:
##  Residual standard error: 55.28 on 17 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0, Adjusted R-Squared: -0.05882
##  Model log likelihood: -87.97 on 16 degrees of freedom (w/Sigma)
##  AIC: 179.9 BIC: 181.7
##
##  Null model: meanstd
##  Null log likelihood: -81.7 on 16 degrees of freedom
##  AIC: 167.4 BIC: 169.2
##  AIC difference (model versus null): -12.53
##  Heuristic Log Bayes Factor (model versus null):  -12.53
```

4.2 Task 3.2 - NAM with Covariates (Gender & HISEI)

```r
attr_full <- read.csv('cleaned_csv/cleaned_attr.csv', sep=";", header=TRUE)
network_w2 <- read.csv('cleaned_csv/cleaned_10_W2.csv', sep=";", header=TRUE, row.names=1)
#specific for classroom '10'
classroom_10_attr <- attr_full[attr_full$classroomID == 10, ]
network_matrix <- as.matrix(network_w2)
student_ids_in_network <- rownames(network_matrix)
attr_matched <- classroom_10_attr[match(student_ids_in_network, as.character(classroom_10_attr$X)), ]
valid_indices <- complete.cases(attr_matched)
network_matrix_clean <- network_matrix[valid_indices, valid_indices]
attr_final <- attr_matched[valid_indices, ]
y <- as.numeric(attr_final$literacy_end)
x_vars <- cbind(
  gender = as.numeric(as.factor(attr_final$gender)),
  hisei = as.numeric(attr_final$HISEI)
)
nam_model_covariates <- lnam(y = y, x = x_vars, W1 = network_matrix_clean)
summary(nam_model_covariates)
```

```
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -30.880 -14.128   2.132  11.163  28.656
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## gender -4.41729    5.89387  -0.749   0.4536
## hisei   1.40125    0.19775   7.086 1.38e-12 ***
## rho1.1 -0.04422    0.02311  -1.914   0.0556 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    14.77      6.426
##
## Goodness-of-Fit:
##  Residual standard error: 17.47 on 14 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.5973, Adjusted R-Squared: 0.511
##  Model log likelihood: -70.02 on 13 degrees of freedom (w/Sigma)
##  AIC: 148 BIC: 151.4
##
##  Null model: meanstd
##  Null log likelihood: -77.42 on 15 degrees of freedom
##  AIC: 158.8 BIC: 160.5
##  AIC difference (model versus null): 10.81
##  Heuristic Log Bayes Factor (model versus null):  9.14
```

4.3 Task 3.3 - Detailed Interpretation

1. SOCIO-ECONOMIC DOMINANCE (HISEI): With an estimate of roughly 1.40 ($p < 0.001$), HISEI is the primary driver of literacy. This suggests that academic outcomes in Classroom 10 are highly stratified by parental background/status rather than social interactions.

2. GENDER NEUTRALITY: Gender ($p = 0.4536$) does not significantly predict literacy. While gender often matters in social structure, it is not an 'academic' divider here.

3. MARGINAL AUTOCORRELATION (rho): The rho value is -0.044 with $p = 0.0556$. While technically not significant at the 5% alpha level, it is "marginally significant" (if we consider it against a more lenient 10%). Interestingly, the negative sign suggests a slight (though unproven) tendency for dissimilarity among neighbors, which contradicts the peer influence hypothesis.

4. MODEL EXPLANATORY POWER: The Multiple R-Squared is 0.597. This indicates that our model accounts for roughly 60% of the variance in literacy scores, which is quite robust. Most of this variance is likely captured by the HISEI variable.

4.4 Task 3.4 - Comparative Reflection (NAM vs. QAP)

By synthesizing Task 2 (MR-QAP) and Task 3 (NAM), we can evaluate the two main social network mechanisms: Selection and Influence.

1. NO EVIDENCE FOR SOCIAL SELECTION (from Task 2): In the QAP, literacy was not a significant predictor of friendship ties. Students are not choosing friends based on similar academic performance.
2. NO EVIDENCE FOR SOCIAL INFLUENCE (from Task 3): The NAM shows rho is not significant. Students are not becoming more similar to their friends over time in terms of literacy.
3. CONTRASTING HOMOPHILY: Gender is the "Social Divider"—it predicts who talks to whom (QAP) but not what they score (NAM). HISEI is the "Academic Divider"—it predicts what they score (NAM) but not who they talk to (QAP).
4. CONCLUSION FOR CLASSROOM 10: Academic achievement in this classroom appears to be an individual attribute rooted in socio-economic background, largely insulated from the social pressures or benefits of the friendship network.

4.5 Task 3.5 - Analysis for all classrooms

We repeat the analysis in (3) for the rest of the classrooms:

```r
library(sna)
library(network)

attr_full <- read.csv('cleaned_csv/cleaned_attr.csv', sep=";", header=TRUE)

classroom_ids <- unique(attr_full$classroomID)

results_list <- list()

pooled_attr <- data.frame()   # ADD THIS LINE

for (cls in classroom_ids) {

  cat("Classroom:", cls, "\n")

  network_file <- paste0('cleaned_csv/cleaned_', sprintf('%02d', cls), '_W2.csv')

  if (!file.exists(network_file)) {
    cat("No Wave 2 network file found for classroom", cls, "- skipping.\n")
    next
  }

  network_w2 <- read.csv(network_file, sep=";", header=TRUE, row.names=1)
  network_matrix <- as.matrix(network_w2)

  classroom_attr <- attr_full[attr_full$classroomID == cls, ]

  student_ids_in_network <- rownames(network_matrix)
  attr_matched <- classroom_attr[match(student_ids_in_network, as.character(classroom_attr$X)), ]

  valid_indices <- complete.cases(attr_matched[, c("literacy_end", "gender", "HISEI")])
```

```r
  if (sum(valid_indices) < 3) {
    cat("Not enough complete cases for classroom", cls, "- skipping.\n")
    next
  }

  network_matrix_clean <- network_matrix[valid_indices, valid_indices]
  attr_final <- attr_matched[valid_indices, ]

  y <- as.numeric(attr_final$literacy_end)
  x_vars <- cbind(
    gender = as.numeric(as.factor(attr_final$gender)),
    hisei  = as.numeric(attr_final$HISEI)
  )

  nam_model <- lnam(y = y, x = x_vars, W1 = network_matrix_clean)

  print(summary(nam_model))

  results_list[[as.character(cls)]] <- nam_model

  attr_final$classroomID_factor <- as.factor(cls)
  pooled_attr <- rbind(pooled_attr, attr_final)
}
```

```
## Classroom: 1
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.206 -18.112   5.482  14.059  33.549
##
## Coefficients:
##         Estimate Std. Error Z value Pr(>|z|)
## gender -4.098403   8.283513  -0.495    0.621
## hisei   0.980383   0.223191   4.393 1.12e-05 ***
## rho1.1  0.008571   0.012577   0.681    0.496
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    19.49      8.255
##
## Goodness-of-Fit:
##  Residual standard error: 20.9 on 20 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.3126, Adjusted R-Squared: 0.2094
##  Model log likelihood: -100.9 on 19 degrees of freedom (w/Sigma)
##  AIC: 209.9 BIC: 214.4
##
##  Null model: meanstd
##  Null log likelihood: -103.6 on 21 degrees of freedom
##  AIC: 211.2 BIC: 213.5
```

```
##  AIC difference (model versus null): 1.336
##  Heuristic Log Bayes Factor (model versus null):  -0.9355
##
## Classroom: 2
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.441 -12.036  -2.089  13.459  65.310
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## gender  8.99504    9.59276   0.938    0.348
## hisei   0.39175    0.25633   1.528    0.126
## rho1.1  0.01655    0.02340   0.707    0.479
##
##       Estimate Std. Error
## Sigma    21.51      11.57
##
## Goodness-of-Fit:
##  Residual standard error: 23.22 on 17 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.1382, Adjusted R-Squared: -0.01385
##  Model log likelihood: -89.78 on 16 degrees of freedom (w/Sigma)
##  AIC: 187.6 BIC: 191.5
##
##  Null model: meanstd
##  Null log likelihood: -87.62 on 18 degrees of freedom
##  AIC: 179.2 BIC: 181.2
##  AIC difference (model versus null): -8.331
##  Heuristic Log Bayes Factor (model versus null):  -10.32
##
## Classroom: 3
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -43.299 -15.330   2.493  23.293  37.291
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## gender -2.71312    6.01882  -0.451  0.65215
## hisei   0.68350    0.21037   3.249  0.00116 **
## rho1.1  0.02676    0.04103   0.652  0.51431
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    22.34      12.49
##
## Goodness-of-Fit:
```

```
##  Residual standard error: 24.55 on 17 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.2779, Adjusted R-Squared: 0.1505
##  Model log likelihood: -90.54 on 16 degrees of freedom (w/Sigma)
##  AIC: 189.1 BIC: 193.1
##
##  Null model: meanstd
##  Null log likelihood: -92.16 on 18 degrees of freedom
##  AIC: 188.3 BIC: 190.3
##  AIC difference (model versus null): -0.7596
##  Heuristic Log Bayes Factor (model versus null):  -2.751
##
## Classroom: 4
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.092 -24.432   4.006  17.924  51.425
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## gender  8.88348   10.20066   0.871  0.38382
## hisei   0.77586    0.24483   3.169  0.00153 **
## rho1.1 -0.01222    0.02540  -0.481  0.63050
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    26.19      19.05
##
## Goodness-of-Fit:
##  Residual standard error: 28.73 on 15 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.1578, Adjusted R-Squared: -0.01061
##  Model log likelihood: -84.32 on 14 degrees of freedom (w/Sigma)
##  AIC: 176.6 BIC: 180.2
##
##  Null model: meanstd
##  Null log likelihood: -84.43 on 16 degrees of freedom
##  AIC: 172.9 BIC: 174.6
##  AIC difference (model versus null): -3.793
##  Heuristic Log Bayes Factor (model versus null):  -5.574
##
## Classroom: 5
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -26.07929 -12.77474   0.02103  15.07141  26.28048
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
```

```
## gender  6.75786    5.45470   1.239    0.215
## hisei   0.66514    0.13634   4.879 1.07e-06 ***
## rho1.1  0.01787    0.01281   1.395    0.163
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    15.93      5.285
##
## Goodness-of-Fit:
##  Residual standard error: 17.03 on 21 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.1634, Adjusted R-Squared: 0.04389
##  Model log likelihood: -100.5 on 20 degrees of freedom (w/Sigma)
##  AIC: 209 BIC: 213.7
##
##  Null model: meanstd
##  Null log likelihood: -97.68 on 22 degrees of freedom
##  AIC: 199.4 BIC: 201.7
##  AIC difference (model versus null): -9.658
##  Heuristic Log Bayes Factor (model versus null):  -12.01
##
## Classroom: 6
## No Wave 2 network file found for classroom 6 - skipping.
## Classroom: 7
## No Wave 2 network file found for classroom 7 - skipping.
## Classroom: 8
## No Wave 2 network file found for classroom 8 - skipping.
## Classroom: 9
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -46.498 -14.093   7.286  20.691  46.040
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## gender 20.30013    8.26517   2.456   0.0140 *
## hisei   0.64696    0.27399   2.361   0.0182 *
## rho1.1 -0.06437    0.04002  -1.609   0.1077
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma     25.2      18.74
##
## Goodness-of-Fit:
##  Residual standard error: 28.8 on 14 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.2194, Adjusted R-Squared: 0.05216
##  Model log likelihood: -79.12 on 13 degrees of freedom (w/Sigma)
##  AIC: 166.2 BIC: 169.6
##
##  Null model: meanstd
```

```
##  Null log likelihood: -79.38 on 15 degrees of freedom
##  AIC: 162.8 BIC: 164.4
##  AIC difference (model versus null): -3.49
##  Heuristic Log Bayes Factor (model versus null):  -5.157
##
## Classroom: 10
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -30.880 -14.128   2.132  11.163  28.656
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## gender -4.41729    5.89387  -0.749   0.4536
## hisei   1.40125    0.19775   7.086 1.38e-12 ***
## rho1.1 -0.04422    0.02311  -1.914   0.0556 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    14.77      6.426
##
## Goodness-of-Fit:
##  Residual standard error: 17.47 on 14 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.5973, Adjusted R-Squared: 0.511
##  Model log likelihood: -70.02 on 13 degrees of freedom (w/Sigma)
##  AIC: 148 BIC: 151.4
##
##  Null model: meanstd
##  Null log likelihood: -77.42 on 15 degrees of freedom
##  AIC: 158.8 BIC: 160.5
##  AIC difference (model versus null): 10.81
##  Heuristic Log Bayes Factor (model versus null):  9.14
##
## Classroom: 11
## No Wave 2 network file found for classroom 11 - skipping.
## Classroom: 12
##
## Call:
## lnam(y = y, x = x_vars, W1 = network_matrix_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -29.339  -7.649   1.745  17.836  23.729
##
## Coefficients:
##        Estimate Std. Error Z value Pr(>|z|)
## gender  8.21419    8.20167   1.002   0.3166
## hisei   0.19101    0.25001   0.764   0.4449
## rho1.1  0.03221    0.01891   1.703   0.0886 .
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##       Estimate Std. Error
## Sigma    15.52      6.708
##
## Goodness-of-Fit:
##  Residual standard error: 17.09 on 15 degrees of freedom (w/o Sigma)
##  Multiple R-Squared: 0.1975, Adjusted R-Squared: 0.03695
##  Model log likelihood: -75.01 on 14 degrees of freedom (w/Sigma)
##  AIC: 158 BIC: 161.6
##
##  Null model: meanstd
##  Null log likelihood: -71.15 on 16 degrees of freedom
##  AIC: 146.3 BIC: 148.1
##  AIC difference (model versus null): -11.73
##  Heuristic Log Bayes Factor (model versus null):  -13.51
```

We summarize the results in a table:

```r
results_table <- data.frame(
  Class = c(1,2,3,4,5,9,10,12),
  Rho_Est = c(0.009, 0.017, 0.027, -0.012, 0.018, -0.064, -0.044, 0.032),
  Rho_p = c(0.496, 0.479, 0.514, 0.631, 0.163, 0.108, 0.056, 0.089),
  Gender_Est = c(-4.10, 8.99, -2.71, 8.88, 6.76, 20.30, -4.42, 8.21),
  Gender_p = c(0.621, 0.348, 0.652, 0.384, 0.215, 0.014, 0.454, 0.317),
  HISEI_Est = c(0.980, 0.392, 0.684, 0.776, 0.665, 0.647, 1.401, 0.191),
  HISEI_p = c(1.12e-05, 0.126, 0.001, 0.002, 1.07e-06, 0.018, 1.38e-12, 0.445),
  R2 = c(0.313, 0.138, 0.278, 0.158, 0.163, 0.219, 0.597, 0.198)
)
knitr::kable(results_table, digits = 3, caption = "Network Autocorrelation Model Results by Classroom")
```

Table 12: Network Autocorrelation Model Results by Classroom

Class	Rho_Est	Rho_p	Gender_Est	Gender_p	HISEI_Est	HISEI_p	R2
1	0.009	0.496	-4.10	0.621	0.980	0.000	0.313
2	0.017	0.479	8.99	0.348	0.392	0.126	0.138
3	0.027	0.514	-2.71	0.652	0.684	0.001	0.278
4	-0.012	0.631	8.88	0.384	0.776	0.002	0.158
5	0.018	0.163	6.76	0.215	0.665	0.000	0.163
9	-0.064	0.108	20.30	0.014	0.647	0.018	0.219
10	-0.044	0.056	-4.42	0.454	1.401	0.000	0.597
12	0.032	0.089	8.21	0.317	0.191	0.445	0.198

We interpret the results:

Classroom 1:
The model estimates a rho of 0.009 with a p-value of 0.496.
The p-value is much greater than 0.05, so we fail to reject the null hypothesis of no autocorrelation. This suggests that there is no significant evidence that students' literacy scores are influenced by those of their friends in this classroom. The gender variable's estimated coefficient is -4.10 This suggests that male students score on average about 4.1 points lower than female students in literacy; however, this coefficient is not statistically significant (p-value = 0.621). In contrast, the HISEI coefficient with an estimate of 0.980 is

highly statistically significant (p-value = 1.12e-05). This indicates that a 1-unit increase in socio-economic status is associates with about 1-point increase in literacy score assuming that all other variables remain constant. The coefficient of determination is $R^2 = 0.313$, meaning that the model explains approximately 31% of the variation in literacy scores, although the adjusted R^2 suggests moderate explanatory power.

Classroom 2:
The model estimates a rho of 0.017 with a p-value of 0.479.
The p-value is much greater than 0.05, so we fail to reject the null hypothesis of no autocorrelation. This suggests that there is no significant evidence that students' literacy scores are influenced by those of their friends in this classroom. The gender variable's estimated coefficient is 8.99 This suggests that male students score on average about 9 points higher than female students in literacy; however, this coefficient is not statistically significant (p-value = 0.348). Similarly, The HISEI coefficient with an estimate of 0.392 is also statistically insignificant (p-value = 0.126), indicating that the socio-economic status does not have a significant association with literacy scores in this classroom. The coefficient of determination is $R^2 = 0.138$, indicating that the model explains about 14% of the variation in literacy scores. The adjusted R^2 is slightly negative, meaning that the explanatory variables provide limited explanatory power.

Classroom 3:
The model estimates a rho of 0.027 with a p-value of 0.514.
The p-value is much greater than 0.05, so we fail to reject the null hypothesis of no autocorrelation. This suggests that there is no significant evidence that students' literacy scores are influenced by those of their friends in this classroom. The gender variable's estimated coefficient is -2.71. This suggests that male students score on average about 2.7 points lower than female students in literacy; however, this coefficient is not statistically significant (p-value = 0.652). In contrast, the HISEI coefficient with an estimate of 0.684 is highly statistically significant (p-value = 0.001). This indicates that a 1-unit increase in socio-economic status is associates with about 0.7-point increase in literacy score assuming that all other variables remain constant. The coefficient of determination is $R^2 = 0.278$, meaning that the model explains approximately 28% of the variation in literacy scores, although the adjusted R^2 suggests moderate explanatory power.

Classroom 4:
The model estimates a rho of -0.012 with a p-value of 0.631.
The p-value is much greater than 0.05, so we fail to reject the null hypothesis of no autocorrelation. This suggests that there is no significant evidence that students' literacy scores are influenced by those of their friends in this classroom. The gender variable's estimated coefficient is 8.88. This suggests that male students score on average about 8.9 points higher than female students in literacy; however, this coefficient is not statistically significant (p-value = 0.384). In contrast, the HISEI coefficient with an estimate of 0.776 is highly statistically significant (p-value = 0.002). This indicates that a 1-unit increase in socio-economic status is associates with about 0.8-point increase in literacy score assuming that all other variables remain constant. The coefficient of determination is $R^2 = 0.158$, indicating that the model explains about 16% of the variation in literacy scores. The adjusted R^2 is slightly negative, meaning that the explanatory variables provide limited explanatory power.

Classroom 5:
The model estimates a rho of 0.018 with a p-value of 0.163.
The p-value is greater than 0.05, so we fail to reject the null hypothesis of no autocorrelation. This suggests that there is no significant evidence that students' literacy scores are influenced by those of their friends in this classroom. The gender variable's estimated coefficient is 6.76. This suggests that male students score on average about 6.8 points higher than female students in literacy; however, this coefficient is not statistically significant (p-value = 0.215). In contrast, the HISEI coefficient with an estimate of 0.665 is highly statistically significant (p-value = 1.07e-06). This indicates that a 1-unit increase in socio-economic status is associates with about 0.67-point increase in literacy score assuming that all other variables remain constant. The coefficient of determination is $R^2 = 0.163$, indicating that the model explains about 16% of the variation in literacy scores, although the adjusted R^2 suggests limited explanatory power.

Classroom 9:
The model estimates a rho of -0.064 with a p-value of 0.108.
The p-value is greater than 0.05, so we fail to reject the null hypothesis of no autocorrelation. However,

the result is marginally close to statistical significance, suggesting weak evidence of a negative network relationship, although it is not statistically significant. The gender coefficient is statistically significant (p-value = 0.014) and the estimated coefficient is 20.30. This suggests that male students score on average about 20 points higher than female students in literacy. The HISEI coefficient with an estimate of 0.647 is also statistically significant (p-value = 0.018). This indicates that a 1-unit increase in socio-economic status is associates with about 0.65-point increase in literacy score assuming that all other variables remain constant. The coefficient of determination is $R^2 = 0.219$, indicating that the model explains about 22% of the variation in literacy scores, although the adjusted R^2 suggests limited explanatory power.

Classroom 12:
The model estimates a rho of 0.032 with a p-value of 0.089.
The p-value is greater than 0.05, but below 0.10, suggesting weak evidence of positive network autocorrelation. This suggests that students' literacy scores may be somewhat positively associated with the literacy scores of their friends, although the evidence is only marginal. The gender variable's estimated coefficient is 8.21. This suggests that male students score on average about 8.2 points higher than female students in literacy; however, this coefficient is not statistically significant (p-value = 0.317). The HISEI coefficient with an estimate of 0.191 is statistically insignificant (p-value = 0.445). This indicates that socio-economic status does not appear to have a significant association with literacy score in this classroom. The coefficient of determination is $R^2 = 0.198$, indicating that the model explains about 20% of the variation in literacy scores, although the adjusted R^2 suggests limited explanatory power.

4.6 Task 3.6 - Interpretation of results

Across all classrooms, there is little evidence which supports the network autocorrelation hypothesis. In most classrooms the estimated rho network autocorrelation parameter is statistically insignificant, indicating that in general, student's literacy scores are generally not strongly associated with the other student's literacy scores. Only a couple classrooms (10 and 12) show some weak evidence of network autocorrelation, where the p-values are close to significance thresholds.

The gender variable only appeared to be statistically significant for Classroom 9, where male students tended to have much higher literacy scores compared to female students. However, gender appeared to be statistically insignificant in the rest of the classrooms.

Socio-economic status on the other hand tended to have a stronger relationship with literacy scores. In most classrooms (1,3,4,5,9 and 10), the HISEI coefficient was statistically significant. This suggests that students with higher socio-economic status tended to achieve higher literacy scores.

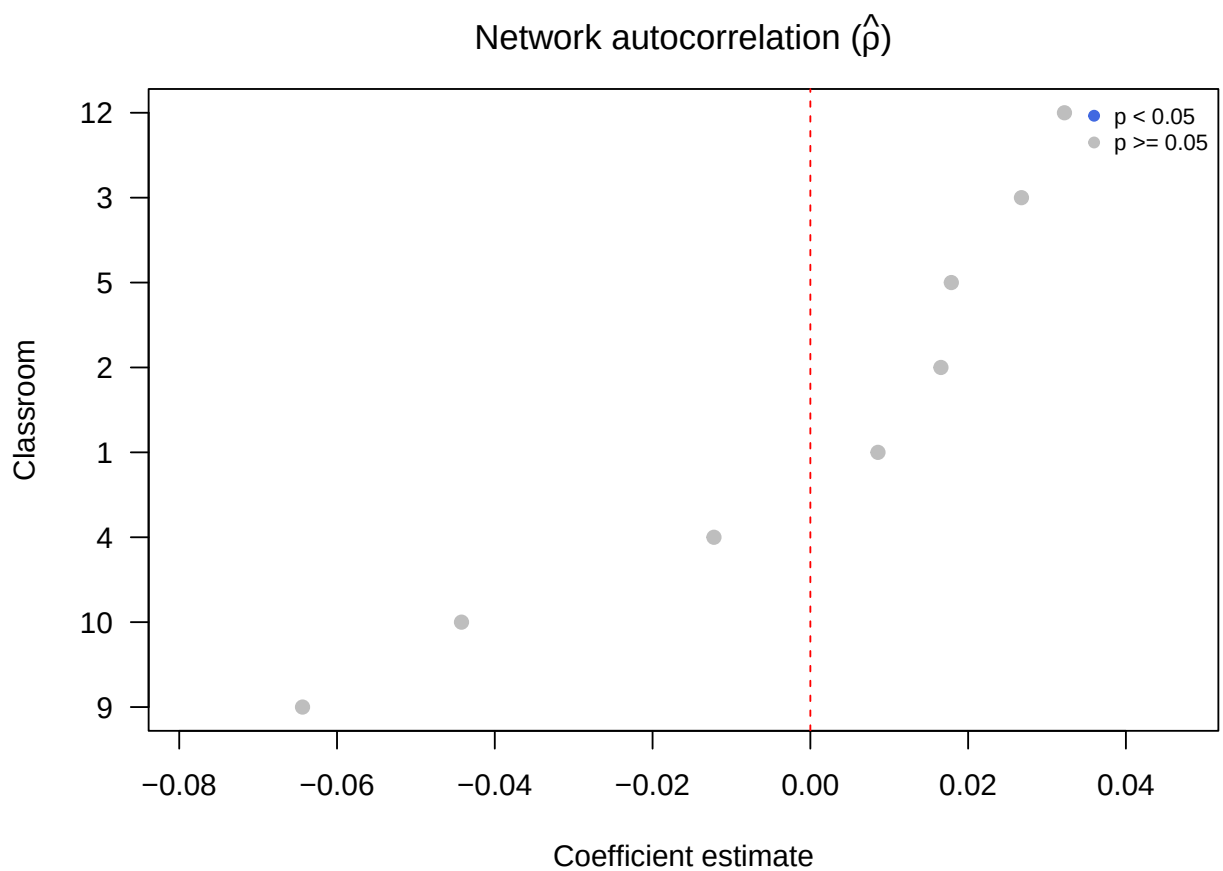
The coefficient of determination R^2 observed across classrooms tended to be low, with Classroom 10 having the largest value of R^2. However, in most classrooms the models seemed to explain only a small proportion of variation in literacy scores, indicating that other factors that were not included in the model may be statistically significant for their effect on literacy scores

Overall, these results suggest that the autocorrelation hypothesis is not strongly supported in this dataset, while other factors such as the socio-economic status appeared to be more significant in explaining literacy scores. These results vary across classrooms and therefore they should be not generalizable to other classrooms or to other populations.

We now visualize the results across classrooms, to summarize the estimated network autocorrelation effects and the relationship between HISEI and literacy scores.

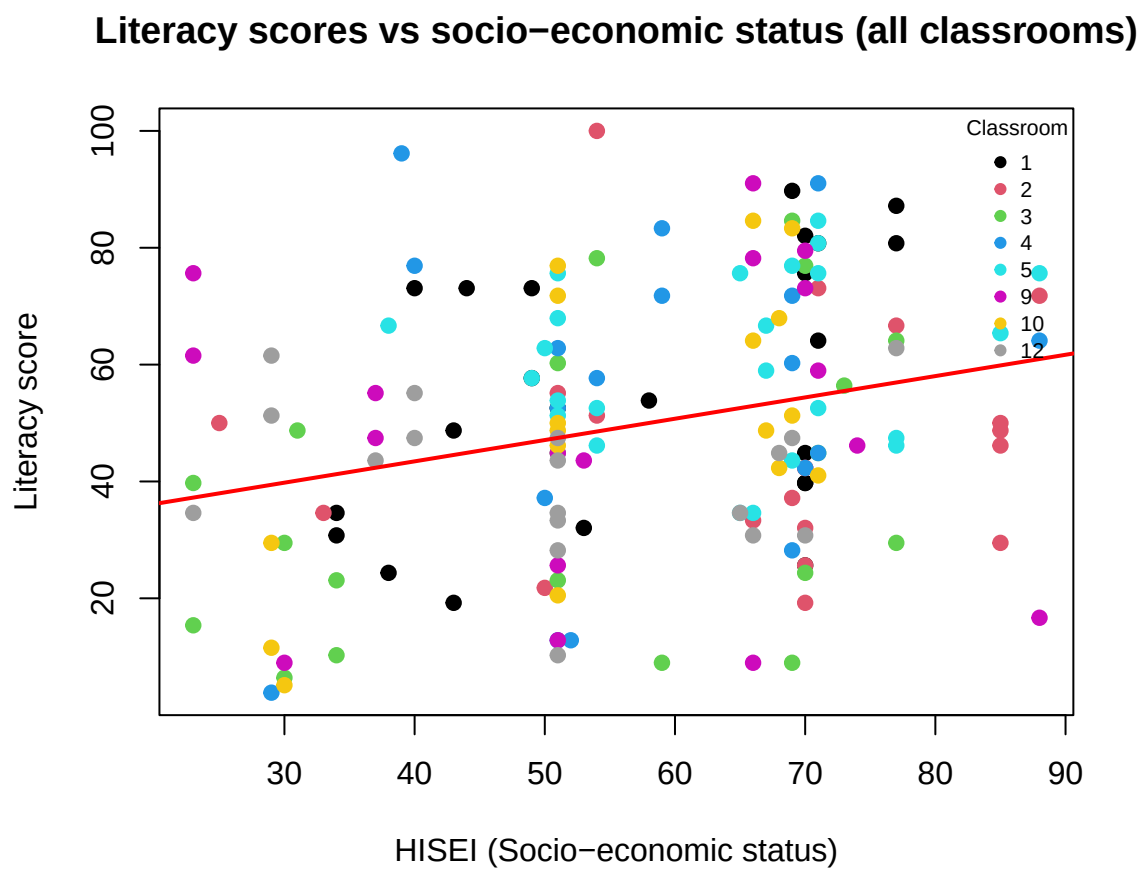
```
results_summary <- data.frame(
  classroom = c(1, 2, 3, 4, 5, 9, 10, 12),
  estimate = c(0.008571, 0.01655, 0.02676, -0.01222, 0.01787, -0.06437, -0.04422, 0.03221),
  p_value = c(0.496, 0.479, 0.514, 0.631, 0.163, 0.1077, 0.0556, 0.0886)
)
```

```r
sub <- results_summary[order(results_summary$estimate), ]

cols <- ifelse(sub$p_value < 0.05, "royalblue", "grey")

rng <- range(sub$estimate)
pad <- diff(rng) * 0.15

par(mar = c(4, 4, 3, 1))

plot(sub$estimate, seq_len(nrow(sub)),
     xlab = "Coefficient estimate",
     ylab = "Classroom",
     main = expression("Network autocorrelation (" * hat(rho) * ")"),
     pch = 19,
     col = cols,
     yaxt = "n",
     xlim = c(rng[1] - pad, rng[2] + pad))

axis(2, at = seq_len(nrow(sub)), labels = sub$classroom, las = 1)
abline(v = 0, lty = 2, col = "red")
legend("topright",
       legend = c("p < 0.05", "p >= 0.05"),
       col = c("royalblue", "grey"),
       pch = 19, cex = 0.75, bty = "n")
```



The caterpillar plot shows that the estimated network autocorrelation coefficients are generally close to zero and not statistically significant. This suggests that friendship networks do not consistently influence literacy scores across classrooms, providing little support for the network autocorrelation hypothesis.

```
plot(pooled_attr$HISEI, pooled_attr$literacy_end,
     xlab = "HISEI (Socio-economic status)",
     ylab = "Literacy score",
     main = "Literacy scores vs socio-economic status (all classrooms)",
     pch = 19,
     col = as.integer(pooled_attr$classroomID_factor))

abline(lm(literacy_end ~ HISEI, data = pooled_attr),
       col = "red",
       lwd = 2)

legend("topright",
       legend = levels(pooled_attr$classroomID_factor),
       col = seq_along(levels(pooled_attr$classroomID_factor)),
       pch = 19, cex = 0.7, title = "Classroom", bty = "n")
```



The scatterplot illustrates the relationship between students' socio-economic status (HISEI) and their literacy scores pooled across all eight classrooms. The fitted regression line has a positive slope, consistent with the finding across individual classrooms that higher HISEI is associated with higher literacy scores. There

is however substantial spread around the line, indicating substantial variability in literacy outcomes that cannot be explained by socio-economic status alone. This aligns with the moderate R^2 values observed in most classrooms. Points are coloured by classroom, and as it appears no single classroom appears to be driving the overall trend.